

Enabling Remote Hand Guidance in Video Calls Using Directional Force Illusion

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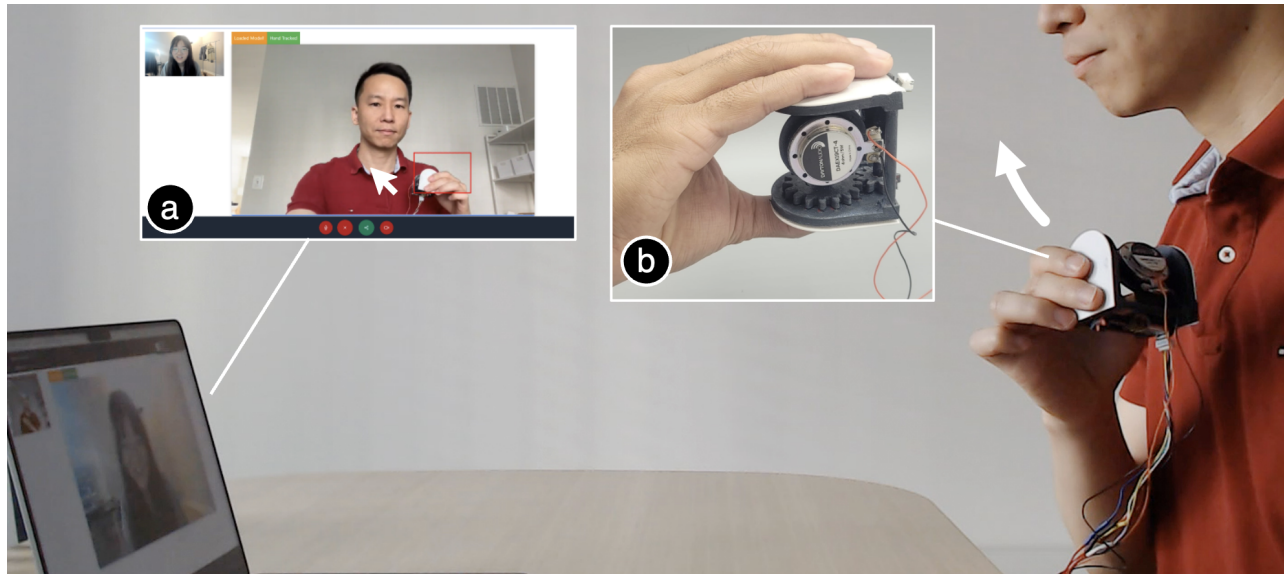


Figure 1: Physicalized remote meeting enabled by (a) a remote meeting system that allows a remote user to move the local user's hand by clicking on the video and (b) a handheld device creating directional force illusion using asymmetric vibration.

ABSTRACT

In current video meetings, communicating positions and hand movements is often challenging, as the meeting participants need to rely on gestures and verbal explanations, resulting in inefficient video meetings. We propose a video meeting system that utilizes a novel directional force feedback method so that people can efficiently guide each other's hand without the need for lengthy verbal explanations. The force feedback method uses asymmetric vibrations to create the illusion of directional pulling forces, which makes it possible to be built as a small and lightweight device. We also developed an online meeting system that allows a user to simply click a location on the remote user's video to move their hand towards the target location.

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1 INTRODUCTION

Online video meeting has become a common form of communication, enabling people to collaborate with others [16, 29, 38], take classes [21, 24, 33], and even get medical care [6, 10, 40]. While high-fidelity video and audio allow rich conversations without physically being in the same place, the lack of a shared physical frame of reference often becomes a cause of friction [7, 16, 30]. During physical encounters, a shared frame of reference makes communicating positions easy by using deictic gestures [2, 15, 29, 41]. In certain in-person meetings, people even hold the other person's hand to guide their hand movement. For example, when learning musical instruments, tutors often correct the student's poses by moving their hands. However, doing so in video meetings often involves repetitive verbal guides (e.g., "to the left") and hand gestures with abstract orientations and directionality, making such interactions difficult and inefficient [18]. This inefficiency may make activities involving physical interactions, such as rehabilitation or engineering collaboration, difficult to be done through video calls.

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Researchers have shown that the use of devices that change position or shape can enhance remote communication, making conversation intimate [9] and providing a better sense of co-presence and more realistic experiences [31]. For example, inFORM [14] supports tangible interaction for video-mediated communication with variable stiffness rendering. Feick et al. [13] proposed using an actuated physical proxy, a robot, to help address perspective challenges in remote communication, enabling bidirectional telemanipulation of physical objects. In enhancing the mobility and physicality of remote audiences, prior work used telepresence robots [32, 42] or swarms of telepresence robots [25] to enable physical embodiments of remote people. To assist remote user studies with more contextual information, Remotion [34] supports replaying user hand movements using a custom-made robotic holder. Despite the benefits of using shared physical control, these methods require large and grounded devices, such as the actuated table [14] and a robotic arm [25], making them difficult to be used at homes and offices.

Smaller and ungrounded force feedback devices can be a more practical choice for enabling hand-guiding interactions in remote communication. For example, body-grounded force feedback devices can create directional force feedback at the user's hand by pushing it against the user's body wearing the device [8, 20]. Airflow can also create directional force by using compressed air [17, 36] or propeller-generated wind [19, 22]. Electrical muscle stimulation (EMS) uses electrodes attached to the user's body to stimulate muscles and move arms and hands [26–28]. Asymmetric vibration based illusion methods use a vibrotactile actuator or an oscillating mass to create the movement with an asymmetric movement profile, producing a sense of pull or push to the fingers holding the device due to the non-linear characteristics of human tactile perception [3, 5, 12, 35]. Among these methods, we found the asymmetric vibration based illusion method to be most promising for video meetings since it can be implemented with small and affordable actuators (i.e., vibrotactile actuators) and does not have the drawbacks of other methods, such as the large size of the device (body-grounded device), excessive wind and noise (airflow-based methods), and the need for attaching electrodes on the user's body (EMS), which may make them unsuitable for everyday video meetings.

In this work, we present a video meeting system that uses a handheld directional force illusion generation device. With the developed system, a user can simply click on the location on the video to move the remote user's hand towards that location. The work has two novel components: 1) the video meeting platform that accommodates user interactions for guiding remote participant's hand movement (Fig. 1a), and 2) the design and implementation of a handheld directional force illusion generation device (Fig. 1b). The handheld device uses the novel design of rotating a voice coil actuator to create a pulling force illusion in arbitrary directions. Our preliminary study on the handheld device performance showed that, when force illusions in eight different directions were given, the participants could correctly recognize 37.3% of them on average and 66.59% if responses with 1-interval errors are considered correct. While the device performance needs improvements, we believe that the use of directional haptic feedback in video meetings will help improve the accessibility of essential services by supporting more physical interactions in online meetings.

2 REMOTE HAND GUIDANCE IN VIDEO CALLS

We envision that users will be able to use an online meeting interface that is similar to conventional online meeting tools, with the addition of a simple click-on-video interaction to guide the remote user's hand. Enabling this requires the design and implementation of two components, the hand-guiding haptic device and remote meeting tool, which we explain here.

2.1 Hand Guidance Device

We set two design goals to ensure the convenient and effective use of the device in video meetings. First, the device should be small and lightweight for the user to hold for video calls, and second, the device should be able to create the force illusion in any direction in 2D space to move the user's hand in regard to the clicked location on the video. While prior research on asymmetric vibration illusion tested various configurations of vibrotactile actuators for a small and lightweight setup [11, 12, 35], these methods do not allow the device to create the force illusion in arbitrary directions. To enable this, we designed a prototype that uses a single vibrotactile actuator that is rotated by a DC motor (See Fig. 2a). The prototype consists of 3D-printed structures, carbon fiber rods, a 5 W Dayton Audio DAEX25CT-4 voice coil actuator, and a micro gear DC motor with an encoder controlled by an Arduino and a TB6612FNG motor driver. The dimensions of the device are 51 mm × 51 mm × 63.5 mm, and it weighs 74 g. The vibration signal was generated by a PC audio interface and amplified by a 40 W Class-AB audio amplifier. We designed the asymmetric vibration driving signal based on a previous design by Tanabe et al. [37]. The signal consists of a 75 Hz asymmetric sine wave with a half cycle inverted for every two cycles (see Fig. 2b). The vibration frequency was heuristically determined through a pilot test. The voice coil actuator can be rotated from 0 to 180 degrees. Combined with the positive and negative asymmetric vibrations that create the opposite directional force illusion, the device can create a directional force illusion in any 2D arbitrary direction. Since the orientation of the device was not measured, the device should be held in the same orientation (see Fig. 1a).

2.2 Remote Meeting System

Our remote meeting system resembles the conventional video meeting systems with a speaker view (Fig. 1a), where the remote user takes a large portion of the window. To allow a local user to guide the remote user's hand to a certain position using our handheld prototype that enables a directional force illusion, we design a user interface with the following workflow. To guide the remote user's hand to a particular position, the local user can simply click on the video. When the local user clicks a location on the video, the click location will be sent to the remote user, which guides the remote user's hand that is holding the device towards the location targeted by the local user. This remote hand guidance is a closed-loop control, with the system continuously monitoring the target location and the actual hand location to calculate the directional vector to control the force feedback. Once the distance between the target and the actual hand position becomes smaller than the threshold (i.e., 100 pixels), the prototype will stop guiding the hand. Moreover, the detected positions of the remote user's hands are visualized as red rectangles (See Fig. 1a) using HandTrack.js [39] to assure that

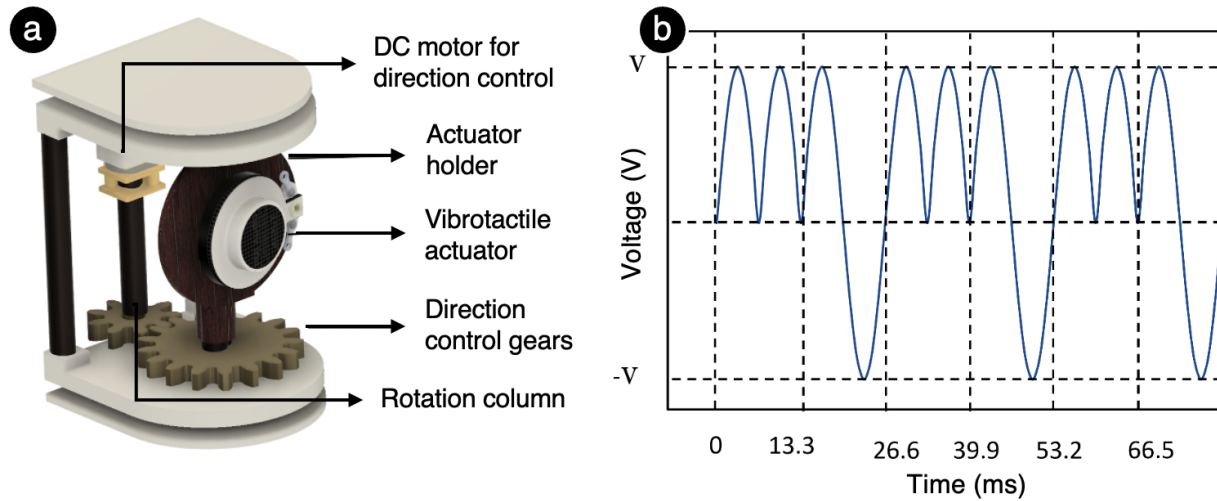


Figure 2: (a) The prototype device design. (b) Asymmetric vibration driving signal waveform.

the local peer can observe the current status of the guidance in real-time. The meeting system is built using Selective Forwarding Unit (SFU) [1] for routing connections, and uses WebSockets to enable communication between front-end and back-end components. The device control driver that handles the connection between the PC and the hand guidance device was written in C# and received the target and hand location data from the web server via a TCP socket.

3 PRELIMINARY STUDY ON THE HAPTIC FEEDBACK

We conducted a preliminary study to test if the illusion of directional force created by the prototype device can be used to provide directional guidance. Eight directional forces separated by 45° intervals were tested in the experiment. In the study, directions 0° , 45° , 90° , 135° utilized the "positive asymmetry" sine wave with half cycle inverted for every two cycles as shown in Fig. 2b to create virtual force and directions 180° , 225° , 270° and 315° utilized "negative asymmetry" sine wave with half cycle inverted for every two cycles. The study protocol was approved by the UVA IRB (#4366) and participants were informed that their participation was voluntary. We recruited 11 participants (9 males and 2 females, 23-32 years old). No participants had experience with asymmetric vibration-based haptic feedback. All participants were right-handed. The study was conducted in a single session that lasted up to one hour. The participants were compensated \$20 for their participation.

3.1 Study setup and task

Participants were seated in front of a 14-inch touchscreen display and wore active noise-canceling headphones playing white noise. The participants were asked to place their dominant hand inside a box covered with black cloth to block device visibility during the study. The position and grip of the prototype were kept consistent across all participants, by holding the device with index fingers on top and thumb on the bottom of the device.

The system played a directional force illusion sample on the prototype device. The order of the force direction was randomized. Each feedback sample, which was a 5-second asymmetric signal, was played twice, with a 2-second interval between them. The participants were asked to identify the direction in which they felt a push or pull by selecting the direction on the touchscreen. There were 10 repetitions for each direction, and there were in total of 80 trials per participant. The session was divided into 5 blocks with a 2-minute break between them, to prevent the participants from getting fatigued.

3.2 Results

Table 1 shows the results of the direction detection study. The participants could correctly identify 8 directions at 37.27% accuracy. If we include responses with 1-interval (45°) error, the correct detection ratio becomes 66.59%. This result may indicate that the participants could gain the approximation of the direction of the force ($\pm 45^\circ$), however, they could not perceive the precise force direction. While the current setup does not provide the most precise directional force illusion, the position-based feedback loop would be able to guide the hand towards the target location.

After completing the tasks, we asked the participants to provide feedback about their experience using Google Forms. When describing their experiences, participants mentioned that they felt a push or pull in their hand, "feels like a push or pull in a certain direction", and a participant mentioned feeling resistance ("Moving my wrist opposite to this direction I felt a resistance.") We also received several comments on the differences in the effectiveness of the force illusion between different directions, for example, "Felt some directions had less pulling force/sensation".

4 DISCUSSION AND FUTURE WORK

We demonstrated the use of a handheld haptic device creating an illusion of directional force to guide hand movements in remote

Direction	0°	45°	90°	135°	180°	225°	270°	315°
Correctness (%)	45.45 (73.64)	47.27 (70)	48.18 (70.91)	31.82 (65.45)	37.27 (55.45)	26.36 (58.18)	35.45 (66.36)	26.36 (72.73)

Table 1: Correctness rates for eight directional forces. The numbers in parentheses indicate the rates including one-interval error.

communication. This illusion-based force feedback method has several advantages over other ungrounded force feedback methods [19, 27, 28, 36], such as its small and lightweight build, quiet operation, and not requiring attaching electrodes. There are a number of potential applications that can benefit from having a remote person actively moving the user's hand, such as remote medical examination where the doctor wants to conduct palpation and onsite workers receiving technical support from a remote expert. We also believe that remote hand guidance will be helpful in improving accessibility, in which remote users with limited vision receive help through hand guidance in video meetings as if they are meeting others in person.

The results of the study also revealed some directions for improvement. First, the study showed that the positive asymmetry vibration (0° - 135° forces) produced more accurate results (15% higher, 43.18% vs. 37.27%). In our study, we only rotated the vibrotactile actuator by 180° as we assumed that inverting the asymmetry would produce similar results. We think that using only the positive asymmetry vibration signal and using the full 360° rotation of the actuator would improve the results. Second, the low direction detection results could be caused by the lack of proper orientational reference on the device, as found in other studies [23], which may become less severe if a physical landmark (e.g., a bump on the device) is given to the user. Third, the asymmetric vibration may have been degraded in the transmission. The vibration created by the voice coil actuator gets delivered to the hand through multiple parts, including the actuator-holding plate, a carbon fiber rod, bearings, and the device cover. A careful vibration profiling with a high-speed displacement sensor may help identify such issues. A prior study also found that actively moving one's hand while feeling the feedback can improve the directional perception of the illusion [4], and we may implement such a protocol for future prototypes to further improve the performance.

In addition to the improvements to the illusory force method, our future work will include a thorough investigation of the usability and effectiveness of the hand-guidance interaction for remote video meetings in various scenarios. It will also investigate the user's perception of such a method, such as the sense of agency while the device generates the hand-guiding force illusion.

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